

Qatar-1b

R-2003

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_230629 · created 2026-08-07T16:10:06+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460124.9746 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.017
Transit depth (Rp/Rs)^2	3.7 +/- 0.65 [%]
Orbital Inclination (inc)	85.12 +/- 1.24
Airmass coefficient 1 (a1)	0.6025 +/- 0.008
Airmass coefficient 2 (a2)	0.42 +/- 0.13
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [332, 155]
Optimal Aperture	3.86
Optimal Annulus	9.91
Transit Duration (day)	0.036 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.017 vs 0.1463 (deviation 2.05 σ)
Duration fit vs published	0.036 d vs 0.0692 d (deviation 7.43 σ)
Mid-time O-C	6.5 min
Depth SNR	8.6
Residual scatter	1.33 %

Light curve

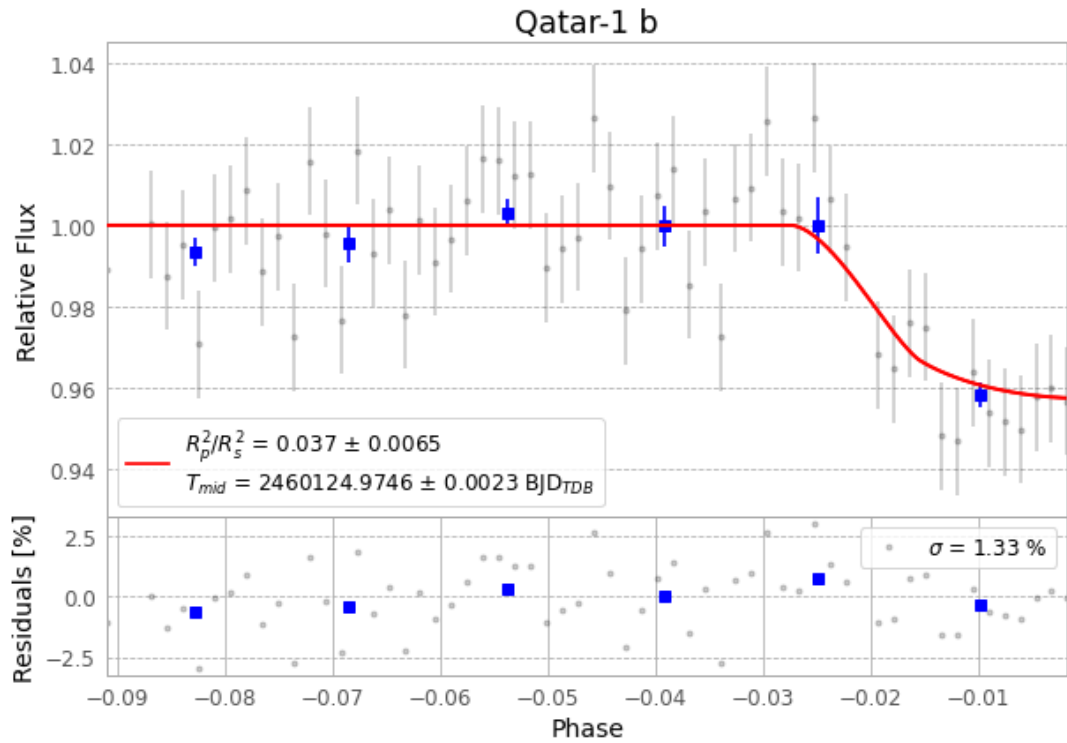


Figure 1 — FinalLightCurve_Qatar-1 b_2023-06-29.png (SHA-256 6dee8cff47417511...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [284, 212], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 f71da9cc99a1b832...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

60 FITS frames (39 MB), Qatar-1230629081547.FITS → Qatar-1230629111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230629.log (6a6f35f1e4fe...), FinalLightCurve_Qatar-1 b_2023-06-29.png (6dee8cff4741...), FinalLightCurve_Qatar-1 b_2023-06-29.csv (a1f28c4b6fc0...)

Compute resources

Wall time 128.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:10Z.